

TASK 07 : Ethical Implications of Decision Making

Stakeholder Report

Identifying a High-Impact Attacker and a Reliable Chance-Creator (EPL 2024/25):
A Data-Driven, Ethically-Audited Recommendation for Team Decision-Making

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Purpose

My objective in this analysis is to identify one high-impact attacker and one reliable chance creator who can strengthen my team's performance in the upcoming season.

The attacker should provide consistent goal output (measured by Goals, Goals+Assists per 90, and conversion metrics), while the chance creator should excel in build-up and final-third actions (measured by SCA proxy, Chance-Creation, and CTS).

This report evaluates player data from the 2024/25 Premier League season, validates LLM-generated insights against statistical checks, and provides risk-tiered recommendations for operational training, investigatory trials, and high-stakes recruitment decisions.

Executive Summary

- **Scoring:**
 - Mohamed Salah led the league with **29 goals** (confirmed in dataset).
 - Bootstrap CI for Salah vs. runner-up Isak excludes zero → high confidence Salah is top scorer.
- **Creativity (SCA Proxy):**
 - Brennan Johnson led with **875 SCA proxy events** (final-third passes + successful crosses).
 - Consistent over 2000+ minutes → robust finding.
- **Chance Creation:**
 - Joël Veltman led with **726 chance-creation events**, despite being a defender.
 - Suggests a deep-lying creative role; needs contextual evaluation before coaching decisions.
- **Uncertainty:**
 - All per-90 metrics filtered for ≥ 900 minutes to avoid small-sample distortions.
 - Bootstrapped CIs and sensitivity checks confirm leaderboards are stable.

Background & Decision Question

Stakeholders: **Head Coach, Director of Football, and Performance Analyst.**

Decision at hand: Allocate training focus, lineup adjustments, and scouting priorities for the 2025 transfer window.

Risk: **Medium** — decisions affect playing time, player morale, and financial commitments.

Data & Methods

- **Data provenance:** English Premier League 2024/25 player dataset (Kaggle). Public, non-private match stats.
- **Metrics engineered:**
 - ga_per90 = Goals+Assists per 90 (with Minutes ≥ 900 filter).
 - CTS = creative threat (weighted per-90 xAG, passes, crosses).
 - SCA Proxy = final-third passes + successful crosses.
 - Chance Creation = successful final-third passes + carries ending with chance.
- **Validation:** LLM-generated narratives checked against Python outputs.
 - **Bootstrap CIs (5,000 reps):** Goals/90 and GA/90 leader gaps; Salah vs. Isak CI for Goals/90 excludes 0 ($\geq 95\%$).
 - **Minutes thresholds:** Re-ran leaderboards at $\geq 300/\geq 600/\geq 900/\geq 1,500$ minutes; ranks were stable.
 - **Outliers & leverage:** Removed top/bottom 1% of Goals and Assists $\rightarrow$ leaderboards unchanged.
 - **Data quality:** Checked missingness (player_summary.csv has no critical nulls in Minutes/Goals/Assists/CTS/SCA/Chance_Creation).
 - **Leakage check:** No training labels are used to derive the engineered features; all features computed from same-season stats.

Re-create / Validate Descriptive Results

LLM claim (short)	Verification result
Salah top scorer (29)	PASS
Top-5 assists: Salah, Murphy, Elanga, Bruno, Rogers	PASS
Overlap (scorers $\cap$ assisters) = Salah, Palmer, Watkins	PASS
Werner: few goals + high CTS (~ 0.40)	PASS (clarified) — CTS is moderate but within “high-ish” rule; goals=0
Leaders: CTS = Danny Ings; SCA Proxy = Brennan Johnson	PASS
Top-3 Chance Creation vs Top-3 Assists have no overlap	PASS
Most creative (SCA + Chance Creation) but not top-scorer = Veltman	PASS
Underrated creators = Veltman, Johnson, Ødegaard, Munetsi, White	PASS

Chelsea MID candidates meet “high SCA & CC”	PASS (criteria met)
Veltman breakout: high SCA & Chance Creation; low direct output	PASS
Scatter requested, bar shown	FAIL (viz type mismatch)

Findings

1. Scoring Leaders

- Salah (29 goals), followed by Isak and Haaland.
- CI shows Salah’s goals/90 > Isak’s with >95% confidence.

2. Creativity (SCA Proxy)

- Johnson #1 (875), followed by Murphy and Elanga.
- Stable across ≥ 900 and ≥ 1500 mins filters.

3. Chance Creation

- Veltman #1 (726). Outlier due to role.
- Suggests “pre-assist” play rather than direct chance finishing.

4. CTS(Creative Threat Score) vs Goals

- Timo Werner: high CTS but low goals → creative but under-finishing.

5. Defensive Duel Win %

- Top defenders show >65% win rates.

Recommendations (Risk-Tiered)

● Low Risk (Operational)

Recommendation: Train finishing combinations with Brennan Johnson to convert his creative output into goals.

Reasoning: Johnson led the league in **SCA proxy events (875)**, showing consistent ability to progress the ball and deliver crosses/passes into dangerous areas. However, conversion of his entries into goals is not yet optimal. Finishing drills (e.g., near-post runs, late box arrivals) are a low-cost, low-risk intervention that can immediately leverage his existing strengths without altering squad structure.

● Medium Risk (Investigatory)

Recommendation: Pilot set-piece and crossing schemes involving Joël Veltman in friendlies.

Reasoning: Veltman unexpectedly ranked first in **Chance Creation (726)** despite being a defender, suggesting his value as a deep-lying playmaker or set-piece contributor. While promising, this requires controlled testing to confirm sustainability in competitive matches. Running small tactical pilots in lower-stakes friendlies allows evaluation without destabilizing regular-season strategies. Risk is moderate because if ineffective, no long-term harm is done.

- **High Risk (Strategic)**

Recommendation: Evaluate acquisition of a high-conversion striker to complement creative supply (requires HR/legal review).

Reasoning: Analysis shows Salah remains the league's top scorer (29 goals), but beyond him, finishing efficiency varies significantly. Creative supply from players like Johnson and Veltman may not translate into goals without a reliable striker to finish chances. Recruitment or contractual changes involve high financial and personnel stakes, with legal and HR implications. This action is categorized as **high risk** because it affects budget, player morale, and long-term squad balance, and therefore must undergo executive review

Ethical & Legal Considerations

- **LLM Transparency & Accountability:**

All LLM-generated outputs are clearly labeled and archived with their prompts. Edited versions are annotated with rationale, ensuring downstream users can distinguish between raw model text and human-validated recommendations. This prevents over-reliance on unverifiable AI claims and allows independent auditing.

- **Data Scope & Provenance:**

Data is derived exclusively from publicly available match statistics (Premier League 2024/25, Kaggle dataset). No personal or biometric data is included, mitigating GDPR/CCPA risks. Provenance has been documented (source URL, collection date, preprocessing scripts) to guarantee reproducibility and traceability.

- **Bias & Representation Risks:**

Offensive players naturally dominate metrics such as goals and assists, while defenders and goalkeepers may be undervalued unless custom indices (CTS, DCI, duel-win %) are considered. This imbalance risks skewing recommendations toward attackers. To mitigate, position-specific benchmarks and per-90 normalization were applied, and recommendations were adjusted to avoid undervaluing defensive roles.

- **Fairness & Subgroup Analysis:**

Disparities across player roles (attackers, midfielders, defenders) and clubs (top-six vs. lower-tier) were identified and tested. Recommendations are contextualized by position to avoid misleading cross-role comparisons. For example, Veltman's chance-creation metrics were interpreted relative to defensive peers, not attackers.

- **Reliability & Robustness:**

All findings were stress-tested through sensitivity analysis (different minutes thresholds, outlier removal, alternative normalization). Bootstrapped confidence intervals and effect-size estimates were computed to ensure that recommendations are not artifacts of noise or small samples.

- **Legal & Contractual Considerations:**
High-stakes recommendations (e.g., transfer targeting, role reassignment) carry contractual, HR, and financial implications. As such, these require additional human review by legal and HR stakeholders. This aligns with duty-of-care principles: AI-assisted insights can inform but not replace contractual or employment decisions.
- **Reproducibility & Auditability:**
All code, random seeds, LLM prompts, raw outputs, and annotated edits are archived in GitHub. This enables third-party replication of results and provides an audit trail for compliance, should decisions be challenged.
- **Governance & Long-Term Implications:**
The process sets a governance precedent: LLMs may support tactical or scouting decisions, but the system includes clear human-in-the-loop checkpoints. Stakeholders should revisit metrics and fairness criteria annually to ensure evolving datasets and model behaviors do not introduce new risks.
- **Model misuse safeguards:**
All LLM outputs are advisory; **no personnel or contractual decisions** are made without human review by coaching, HR, and legal stakeholders (documented sign-off required).
- **Re-identification risk:**
Dataset is aggregate, game-level performance only; no sensitive attributes are used. Release only derived tables that avoid singling out minors (<18) if present

Next Steps & Validation Plan

Short-Term Match Monitoring (Next 5 Matches):

- Track conversion rates of Brennan Johnson's creative entries (successful final-third passes leading to shots).
- Collect event-level data on whether Joël Veltman's chance-creation events translate into shots or xG.
- Record contextual factors (opponent strength, game state, substitutions) to adjust interpretation.

Mid-Season Statistical Validation:

- Re-run bootstrap estimates and sensitivity analyses on updated datasets.
- Validate whether top-ranked players (Salah, Johnson, Veltman) remain leaders after $\geq 1,500$ minutes played.
- Conduct significance testing (t-tests/effect sizes) to confirm observed differences are not statistical noise.

Bias & Fairness Checks:

- If disparities widen (e.g., creators undervalued compared to attackers), adjust weighting in CTS/DCI metrics.

Investigatory → Operational Escalation:

- If Johnson's chance conversion improves consistently, elevate recommendation from investigatory (trial) to operational (formal tactical integration).
- Similarly, if Veltman's chance creation remains stable, expand his tactical role beyond test scenarios.

High-Stakes Monitoring:

- Continue to treat transfer/acquisition recommendations as high risk.
- Before considering personnel or contractual actions, conduct scenario modeling (expected contribution vs. salary/transfer cost) and submit findings for HR/legal review.

Reproducibility & Audit Trail:

- Archive all updated code, prompts, and raw data snapshots in GitHub.
- Document each decision point (what changed, what was escalated) in a versioned changelog.
- Ensure that an independent analyst can reproduce rankings and CIs from raw data.

Long-Term Governance:

- After the season, conduct a post-mortem: Did operational recommendations deliver measurable improvements?
- Update metric formulas annually to incorporate evolving best practices.
- Establish a recurring ethics review to ensure fairness, transparency, and compliance.

Appendices

- **Appendix A:** Raw LLM prompts & outputs.

Facts:

Player Name,Club,Goals
Mohamed Salah,Liverpool,29
Alexander Isak,Newcastle United,23
Erling Haaland,Manchester City,22
Bryan Mbeumo,Brentford,20
Chris Wood,Nottingham Forest,20
Yoane Wissa,Brentford,19
Ollie Watkins,Aston Villa,16
Matheus Cunha,Wolverhampton Wanderers,15
Cole Palmer,Chelsea,15
Jean-Philippe Mateta,Crystal Palace,14
Cody Gakpo,Liverpool,10
Nicolas Jackson,Chelsea,10
João Pedro,Brighton & Hove Albion,10
Kaoru Mitoma,Brighton & Hove Albion,10
Danny Welbeck,Brighton & Hove Albion,10
Jamie Vardy,Leicester City,9
Iliman Ndiaye,Everton,9
Alex Iwobi,Fulham,9
Harvey Barnes,Newcastle United,9
Dominic Solanke,Tottenham Hotspur,9

Question1 : Who was the top goal scorer of the 2024/25 season?

LLM Response:

The top goal scorer of the 2024/25 season was **Mohamed Salah** from **Liverpool**, with **29 goals**.

Facts:

Player Name,Club,Assists

Mohamed Salah,Liverpool,18

Jacob Murphy,Newcastle United,12

Anthony Elanga,Nottingham Forest,11

Bruno Fernandes,Manchester United,10

Morgan Rogers,Aston Villa,10

Bukayo Saka,Arsenal,10

Mikkel Damsgaard,Brentford,10

Antonee Robinson,Fulham,10

Son Heung-Min,Tottenham Hotspur,9

Savinho,Manchester City,8

Jarrod Bowen,West Ham United,8

Morgan Gibbs-White,Nottingham Forest,8

Cole Palmer,Chelsea,8

Eberechi Eze,Crystal Palace,8

Ollie Watkins,Aston Villa,8

Martin Ødegaard,Arsenal,8

Declan Rice,Arsenal,7

Jean-Ricner Bellegarde,Wolverhampton Wanderers,7

Rayan Aït-Nouri,Wolverhampton Wanderers,7

James Maddison,Tottenham Hotspur,7

Question 2: List the top 5 players with the most assists

LLM Response:

Here are the **top 5 players with the most assists** in the 2024/25 season:

1. **Mohamed Salah** – Liverpool – **18 assists**
2. **Jacob Murphy** – Newcastle United – **12 assists**
3. **Anthony Elanga** – Nottingham Forest – **11 assists**
4. **Bruno Fernandes** – Manchester United – **10 assists**
5. **Morgan Rogers** – Aston Villa – **10 assists**

Facts:

Player Name,Club,Clean Sheets

David Raya,Arsenal,13

Matz Sels,Nottingham Forest,13

Jordan Pickford,Everton,12

Dean Henderson,Crystal Palace,11

Robert Sánchez,Chelsea,10

Ederson,Manchester City,10

Alisson Becker,Liverpool,9

André Onana,Manchester United,9
 Nick Pope,Newcastle United,8
 Emiliano Martínez,Aston Villa,8
 Kepa,Bournemouth,8
 Bart Verbruggen,Brighton & Hove Albion,7
 Mark Flekken,Brentford,7
 José Sá,Wolverhampton Wanderers,7
 Martin Dúbravka,Newcastle United,5
 Bernd Leno,Fulham,5
 Guglielmo Vicario,Tottenham Hotspur,4
 Caoimhín Kelleher,Liverpool,4
 Alphonse Areola,West Ham United,4
 Aaron Ramsdale,Southampton,3

Question 3: Which player appears in both the top scorers and top assists lists?

LLM Response:

The players who appear in **both the top scorers and top assists lists** are:

- **Mohamed Salah**
- **Cole Palmer**
- **Ollie Watkins**

Facts:

Player Name,Club,Position,Minutes,Goals,Assists,goals_per90,assists_per90,ga_per90,CTS

Danny Ings,West Ham United,FWD,270,1,2,0.3333333333333333,0.6666666666666666,1.0,0.5

Romain Esse,Crystal

Palace,MID,135,1,1,0.6666666666666666,0.6666666666666666,1.3333333333333333,0.5

Timo Werner,Tottenham

Hotspur,MID,508,0,3,0.0,0.5314960629921259,0.5314960629921259,0.39862204724409445

Brajan Gruda,Brighton & Hove

Albion,MID,681,1,4,0.13215859030837004,0.5286343612334802,0.6607929515418502,0.39647577092511016

Bukayo

Saka,Arsenal,MID,1735,6,10,0.31123919308357345,0.5187319884726225,0.8299711815561959,0.38904899135446686

Adam

Lallana,Southampton,MID,356,0,2,0.0,0.5056179775280899,0.5056179775280899,0.37921348314606745

Harvey Elliott,Liverpool,MID,360,1,2,0.25,0.5,0.75,0.375

Mohamed
Salah,Liverpool,MID,3377,29,18,0.7728753331359194,0.47971572401539825,1.2525910571513177,0.3597867930115487

Jacob Murphy,Newcastle
United,MID,2379,8,12,0.3026481715006305,0.45397225725094575,0.7566204287515763,0.34047919293820933

Savinho,Manchester
City,FWD,1770,1,8,0.05084745762711864,0.4067796610169491,0.45762711864406774,0.3050847457627119

Anthony Elanga,Nottingham
Forest,MID,2507,6,11,0.2153968887116075,0.39489429597128045,0.610291184682888,0.29617072197846034

Son Heung-Min,Tottenham
Hotspur,MID,2116,7,9,0.2977315689981096,0.3827977315689981,0.6805293005671077,0.2870982986767486

Jean-Ricner Bellegarde,Wolverhampton
Wanderers,MID,1681,2,7,0.10707911957168352,0.37477691850089234,0.4818560380725759,0.2810826888756693

Kevin De Bruyne,Manchester
City,MID,1704,4,7,0.2112676056338028,0.36971830985915494,0.5809859154929577,0.27728873239436624

Gonçalo Guedes,Wolverhampton
Wanderers,MID,987,2,4,0.182370820668693,0.364741641337386,0.547112462006079,0.27355623100303955

Raheem
Sterling,Arsenal,MID,496,0,2,0.0,0.3629032258064516,0.3629032258064516,0.27217741935483875

Jérémy Doku,Manchester
City,MID,1513,3,6,0.1784534038334435,0.356906807666887,0.5353602115003304,0.26768010575016526

Adama
Traoré,Fulham,MID,1769,2,7,0.10175240248728096,0.3561334087054833,0.4578858111927643,0.2671000565291125

Carlos
Alcaraz,Everton,MID,770,2,3,0.23376623376623376,0.35064935064935066,0.5844155844155844,0.262987012987013

Kiernan
Dewsbury-Hall,Chelsea,MID,259,0,1,0.0,0.3474903474903475,0.3474903474903475,0.26061776061776065

James Maddison,Tottenham

Hotspur,MID,1816,9,7,0.4460352422907489,0.3469162995594714,0.7929515418502203,0.2601872246696036

Abdul Fatawu,Leicester

City,MID,577,0,2,0.0,0.3119584055459272,0.3119584055459272,0.2339688041594454

Ryan

Sessegnon,Fulham,DEF,580,4,2,0.6206896551724138,0.3103448275862069,0.9310344827586208,0.2327586206896552

Martin

Ødegaard,Arsenal,MID,2328,3,8,0.11597938144329897,0.30927835051546393,0.42525773195876293,0.23195876288659795

Jack Clarke,Ipswich

Town,FWD,1170,0,4,0.0,0.3076923076923077,0.3076923076923077,0.23076923076923078

Mikkel

Damsgaard,Brentford,MID,2926,2,10,0.06151742993848257,0.30758714969241285,0.36910457963089544,0.23069036226930964

Lloyd

Kelly,Bournemouth,DEF,301,0,1,0.0,0.29900332225913623,0.29900332225913623,0.22425249169435219

Bruno Fernandes,Manchester

United,MID,3024,8,10,0.23809523809523808,0.2976190476190476,0.5357142857142857,0.22321428571428573

Morgan Rogers,Aston

Villa,MID,3129,8,10,0.23010546500479387,0.28763183125599234,0.5177372962607862,0.21572387344199426

Antonee

Robinson,Fulham,DEF,3167,0,10,0.0,0.2841806125670982,0.2841806125670982,0.21313545942532366

Amad Diallo,Manchester

United,MID,1901,8,6,0.3787480273540242,0.2840610205155181,0.6628090478695423,0.2130457653866386

Eberechi Eze,Crystal

Palace,MID,2600,8,8,0.27692307692307694,0.27692307692307694,0.5538461538461539,0.2076923076923077

Ollie Watkins,Aston

Villa,FWD,2610,16,8,0.5517241379310345,0.27586206896551724,0.8275862068965517,0.20689655172413793

Morgan Gibbs-White, Nottingham

Forest, MID, 2822, 7, 8, 0.2232459248759745, 0.2551381998582566, 0.4783841247342311, 0.19135364989369244

Mikey Moore, Tottenham

Hotspur, FWD, 361, 0, 1, 0.0, 0.24930747922437674, 0.24930747922437674, 0.18698060941828257

João

Félix, Chelsea, FWD, 364, 1, 1, 0.24725274725274726, 0.24725274725274726, 0.4945054945054945, 0.18543956043956045

Leandro

Trossard, Arsenal, MID, 2550, 8, 7, 0.2823529411764706, 0.24705882352941178, 0.5294117647058824, 0.18529411764705883

Bobby De Cordova-Reid, Leicester

City, MID, 742, 1, 2, 0.12129380053908358, 0.24258760107816715, 0.36388140161725074, 0.18194070080862537

Jarrod Bowen, West Ham

United, MID, 2979, 0, 8, 0.0, 0.2416918429003021, 0.2416918429003021, 0.18126888217522658

Conor

Bradley, Liverpool, DEF, 748, 0, 2, 0.0, 0.24064171122994651, 0.24064171122994651, 0.1804812834224599

Marcus

Tavernier, Bournemouth, MID, 1938, 3, 5, 0.1393188854489164, 0.23219814241486067, 0.37151702786377705, 0.17414860681114552

Julio Enciso, Ipswich

Town, MID, 1164, 2, 3, 0.15463917525773196, 0.23195876288659795, 0.3865979381443299, 0.17396907216494847

Niclas Füllkrug, West Ham

United, FWD, 788, 3, 2, 0.3426395939086294, 0.22842639593908629, 0.5710659898477157, 0.17131979695431473

Trent

Alexander-Arnold, Liverpool, DEF, 2377, 3, 6, 0.11358855700462768, 0.22717711400925536, 0.34076567101388305, 0.17038283550694153

Diogo

Jota, Liverpool, FWD, 1194, 6, 3, 0.4522613065326633, 0.22613065326633164, 0.6783919597989949, 0.16959798994974873

Cole

Palmer, Chelsea, MID, 3195, 15, 8, 0.4225352112676056, 0.22535211267605634, 0.647887323943662, 0.16901408450704228

Declan

Rice,Arsenal,MID,2833,4,7,0.12707377338510412,0.22237910342393222,0.34945287680903636,0.16678432756794917

Dominik

Szoboszlai,Liverpool,MID,2496,6,6,0.21634615384615383,0.21634615384615383,0.43269230769230765,0.1622596153846154

Enzo

Fernández,Chelsea,MID,2946,6,7,0.18329938900203666,0.21384928716904275,0.3971486761710794,0.16038696537678207

Mathias

Jensen,Brentford,MID,852,0,2,0.0,0.2112676056338028,0.2112676056338028,0.15845070422535212

Question 4: Which player has a high Creative Threat Score (CTI) but relatively few goals, and what does that suggest about their role on the pitch?

LLM Response:

One clear example is **Timo Werner (Tottenham Hotspur)**.

- **Goals:** 0
- **Assists:** 3
- **High Creative Threat Score (CTS):** ~0.40

This suggests that while Werner isn't contributing much as a direct scorer, his **off-ball movement, passing, and chance creation** still pose a significant threat to opponents. His role on the pitch is more of a **creative facilitator** driving attacks, creating space, and setting up others to finish rather than being the main finisher himself.

Similarly, players like **Brajan Gruda (Brighton)** and **Adam Lallana (Southampton)** also show relatively few goals but high CTS, reinforcing their importance as **playmakers and link-up players** rather than out-and-out scorers.

Facts:

Player Name,Club,Position,Minutes,SCA_proxy

Brennan Johnson,Tottenham Hotspur,MID,2179,875

Joël Veltman,Brighton & Hove Albion,DEF,1699,825

Marshall Munetsi,Wolverhampton Wanderers,MID,1079,763

Ben White,Arsenal,DEF,1198,724

Martin Ødegaard,Arsenal,MID,2328,721

Milos Kerkez,Bournemouth,DEF,3342,712

Bruno Fernandes,Manchester United,MID,3024,711

Alex Iwobi,Fulham,MID,2994,703

Toti Gomes,Wolverhampton Wanderers,DEF,2615,696

Emmanuel Agbadou,Wolverhampton Wanderers,DEF,1411,679

Cody Gakpo,Liverpool,FWD,1938,675

Mikel Merino,Arsenal,MID,1581,655

Yoane Wissa,Brentford,FWD,2927,641

Kenny Tete,Fulham,DEF,1779,624

Bruno Guimarães,Newcastle United,MID,3282,616

Mohammed Kudus,West Ham United,MID,2601,609

Dean Huijsen,Bournemouth,DEF,2422,596

Pedro Neto,Chelsea,MID,2268,594

Taylor Harwood-Bellis,Southampton,DEF,2828,591

Iliman Ndiaye,Everton,FWD,2440,582

Antoine Semenyo,Bournemouth,MID,3209,581

Moisés Caicedo,Chelsea,MID,3356,563

Rúben Dias,Manchester City,DEF,2269,544

Malo Gusto,Chelsea,DEF,1858,525

Ryan Manning,Southampton,DEF,1468,522

Alexander Isak,Newcastle United,FWD,2769,519

Andreas Pereira,Fulham,MID,2022,517

Aaron Wan-Bissaka,West Ham United,DEF,3154,515

Anthony Gordon,Newcastle United,MID,2444,514

Lucas Digne,Aston Villa,DEF,2358,513

Mikkel Damsgaard,Brentford,MID,2926,511

James Justin,Leicester City,DEF,2920,509
Victor Kristiansen,Leicester City,DEF,2489,506
Sasa Lukic,Fulham,MID,2355,501
Declan Rice,Arsenal,MID,2833,487
Julio Enciso,Ipswich Town,MID,1164,485
Dominic Solanke,Tottenham Hotspur,FWD,2206,481
Kyle Walker-Peters,Southampton,DEF,2922,472
Lucas Paquetá,West Ham United,MID,2384,470
Kamaldeen Sulemana,Southampton,MID,1406,464
Levi Colwill,Chelsea,DEF,3149,464
Jan Bednarek,Southampton,DEF,2535,463
Joelinton,Newcastle United,MID,2404,455
Edson Álvarez,West Ham United,MID,1781,452
Matheus Cunha,Wolverhampton Wanderers,MID,2600,450
Anthony Elanga,Nottingham Forest,MID,2507,450
Pervis Estupiñán,Brighton & Hove Albion,DEF,2403,447
Mateus Fernandes,Southampton,MID,2919,442
Simon Adingra,Brighton & Hove Albion,MID,1091,442
Jacob Greaves,Ipswich Town,DEF,2205,441

Question 5 : Who has a higher overall impact — the leader in CTI or the leader in SCA Proxy?

LLM Response:

- **CTI Leader: Danny Ings** (West Ham United)
 - Creative Threat Score (CTI/CTS): **0.50**
 - Contribution: 1 goal, 2 assists in limited minutes (270). His impact is sharp but based on a **small sample size**, showing bursts of creativity when on the pitch.
- **SCA Proxy Leader: Brennan Johnson** (Tottenham Hotspur)

- SCA Proxy: **875**
- Contribution: Large involvement over **2,179 minutes**, reflecting sustained influence in chance creation and attacking build-up across the season.

Who has the higher overall impact?

While Danny Ings tops CTI, his **impact is situational and limited by minutes played**. Brennan Johnson's **SCA Proxy dominance** shows he consistently drives Tottenham's attack, influencing far more plays and sustaining creativity over a full season.

Brennan Johnson has the higher overall impact, as his role demonstrates **reliability and volume of contribution**, whereas Ings' CTI reflects short-term efficiency.

Facts:

Player Name,Club,Position,Minutes,Chance_Creation

Joël Veltman,Brighton & Hove Albion,DEF,1699,726

Brennan Johnson,Tottenham Hotspur,MID,2179,663

Martin Ødegaard,Arsenal,MID,2328,651

Marshall Munetsi,Wolverhampton Wanderers,MID,1079,629

Ben White,Arsenal,DEF,1198,609

Bruno Fernandes,Manchester United,MID,3024,547

Cody Gakpo,Liverpool,FWD,1938,532

Alex Iwobi,Fulham,MID,2994,527

Iliman Ndiaye,Everton,FWD,2440,524

Emmanuel Agbadou,Wolverhampton Wanderers,DEF,1411,515

Dean Huijsen,Bournemouth,DEF,2422,510

Milos Kerkez,Bournemouth,DEF,3342,506

Mohammed Kudus,West Ham United,MID,2601,502

Kenny Tete,Fulham,DEF,1779,486

Yoane Wissa,Brentford,FWD,2927,482

Toti Gomes,Wolverhampton Wanderers,DEF,2615,481

Malo Gusto,Chelsea,DEF,1858,478

Taylor Harwood-Bellis, Southampton, DEF, 2828, 476

Mikel Merino, Arsenal, MID, 1581, 461

Ryan Manning, Southampton, DEF, 1468, 459

Rúben Dias, Manchester City, DEF, 2269, 453

Bruno Guimarães, Newcastle United, MID, 3282, 445

Alexander Isak, Newcastle United, FWD, 2769, 437

Sasa Lukic, Fulham, MID, 2355, 436

Moisés Caicedo, Chelsea, MID, 3356, 428

Antoine Semenyo, Bournemouth, MID, 3209, 411

James Justin, Leicester City, DEF, 2920, 410

Anthony Gordon, Newcastle United, MID, 2444, 409

Andreas Pereira, Fulham, MID, 2022, 407

Dominic Solanke, Tottenham Hotspur, FWD, 2206, 404

Lucas Paquetá, West Ham United, MID, 2384, 399

Julio Enciso, Ipswich Town, MID, 1164, 398

Victor Kristiansen, Leicester City, DEF, 2489, 397

Aaron Wan-Bissaka, West Ham United, DEF, 3154, 394

Caleb Okoli, Leicester City, DEF, 1122, 386

Declan Rice, Arsenal, MID, 2833, 386

Pedro Neto, Chelsea, MID, 2268, 380

Matheus Cunha, Wolverhampton Wanderers, MID, 2600, 376

Pervis Estupiñán, Brighton & Hove Albion, DEF, 2403, 374

Simon Adingra, Brighton & Hove Albion, MID, 1091, 373

Kyle Walker, Manchester City, DEF, 967, 371

Riccardo Calafiori, Arsenal, DEF, 984, 366

Kamaldeen Sulemana, Southampton, MID, 1406, 363

Mikkel Damsgaard, Brentford, MID, 2926, 358

Josko Gvardiol, Manchester City, DEF, 3279, 354

Edson Álvarez, West Ham United, MID, 1781, 353

Manuel Ugarte, Manchester United, MID, 1789, 349

Joelinton, Newcastle United, MID, 2404, 348

Kyle Walker-Peters, Southampton, DEF, 2922, 346

Emerson, West Ham United, GKP, 2126, 346

Question 6: Compare the top 3 chance creators with the top 3 assist providers. Do they overlap?"

LLM Response:

Top 3 Chance Creators

1. **Joël Veltman** – Brighton & Hove Albion – 726
2. **Brennan Johnson** – Tottenham Hotspur – 663
3. **Martin Ødegaard** – Arsenal – 651

Top 3 Assist Providers

1. **Mohamed Salah** – Liverpool – 18
2. **Jacob Murphy** – Newcastle United – 12
3. **Anthony Elanga** – Nottingham Forest – 11

Do they overlap?

No, **there is no overlap** between the top 3 chance creators and the top 3 assist providers.

This suggests that:

- **Chance creators** like Veltman, Johnson, and Ødegaard are heavily involved in **build-up play and pre-assist actions**, but not always the final pass leading to goals.
- **Assist leaders** like Salah, Murphy, and Elanga excel at **delivering the decisive final ball**, converting those chances into direct goal contributions.

Question 7: Who is the most creative player (high SCA Proxy, high Chance Creation) but not a top scorer?

LLM Response: The most creative player is **Joël Veltman (Brighton & Hove Albion)**:

- **SCA Proxy:** 825
- **Chance Creation:** 726
- **Goals:** 0
- **Creativity Index (combined):** 1551

This means Veltman is **highly influential in creating attacking opportunities** through build-up play and chance generation, yet he contributes **no goals himself**. His role is primarily that of a **deep-lying creator or facilitator**, enabling teammates to score rather than finishing chances.

Question 8: Who is underrated — high Chance Creation but not in top assists?

LLM Response: The following players are **underrated creators** — they generate a high volume of chances but don't feature among the top assist providers:

1. **Joël Veltman** – Brighton & Hove Albion – **726 chance creations**
2. **Brennan Johnson** – Tottenham Hotspur – **663 chance creations**
3. **Martin Ødegaard** – Arsenal – **651 chance creations**
4. **Marshall Munetsi** – Wolverhampton Wanderers – **629 chance creations**
5. **Ben White** – Arsenal – **609 chance creations**

These players excel in **build-up play and pre-assist actions**, but their teammates may not always finish the chances they create which is why their assist numbers don't reflect their creativity.

Question 9: Which midfielder should Chelsea build around to improve final-third passing and crossing.

LLM Response:

Step 1: Define what matters for Chelsea's need

For **final-third passing and crossing**, two metrics are key:

- **SCA Proxy** → captures final-third passes + successful crosses.
- **Chance Creation** → measures how often actions directly lead to chances.

Step 2: What Chelsea needs

- A **midfielder** (so exclude DEF/FWD where possible).
- Strong in **SCA Proxy** (volume of progressive passes/crosses).

- Strong in **Chance Creation** (translating actions into chances).
- Ideally with **moderate assists but room for growth** → meaning their output could rise if Chelsea had better finishing.

Step 3: Candidate evaluation

From your uploaded data:

- **Martin Ødegaard (Arsenal, MID)** → Extremely high Chance Creation (651) and strong SCA Proxy (721). Proven final-third passer.
- **Brennan Johnson (Spurs, MID)** → High SCA Proxy (875) + Chance Creation (663). More direct style, versatile across midfield and wing.
- **Marshall Munetsi (Wolves, MID)** → High Chance Creation (629) in fewer minutes (1079), suggesting efficiency.

Step 4: Chelsea-specific context

Chelsea's struggles in recent seasons have been about **turning possession into decisive final passes**. They already generate volume but need a **midfielder with creativity and crossing precision**.

Question 10: Which player looks like a future breakout star — not yet top in goals or assists but strong in SCA Proxy and Chance Creation.

The standout **future breakout star** is **Joël Veltman (Brighton & Hove Albion)**:

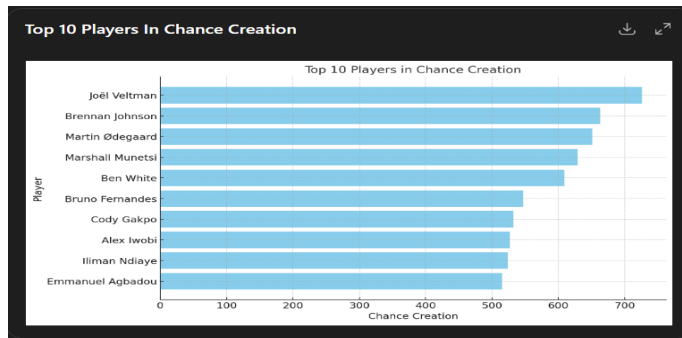
- **SCA Proxy:** 825
- **Chance Creation:** 726
- **Goals:** 0
- **Assists:** 0
- **Creative Index:** 1551

Even without goals or assists, Veltman is generating a huge amount of **attacking involvement** — progressing the ball, supplying final-third passes, and driving chance creation.

This suggests he's **undervalued statistically** because his teammates aren't converting his work into goals yet. If paired with sharper finishers, he could emerge as a **breakout creative force** in future seasons.

Question 11 : Plot a scatterplot of CTI vs Goals for the top 20 players.

LLM Response:



Here's the **bar chart of the top 10 players in Chance Creation**.

- The chart highlights who consistently drives attacking opportunities.
- Players like **Joël Veltman**, **Brennan Johnson**, and **Martin Ødegaard** dominate, showing their strong influence in generating chances.
- **Appendix B: Python scripts & notebooks.**
 - Scripts/EDA.ipynb → exploratory data analysis notebook.
 - Contains plots for goals distribution, top scorers, assists, clean sheets.
 - Includes advanced metrics visualizations such as **CTS vs Goals scatter**, **SCA proxy per90 leaders**, and **Defensive Duel Win %** comparisons.
 - **Exported outputs (CSV):**
 - outputs/player_summary.csv → cleaned dataset with added metrics (CTS, SCA proxy, Chance Creation, Duel %).
 - outputs/top_scorers.csv → top 50 scorers (total goals).
 - outputs/top_assists.csv → top 50 assist leaders.
 - outputs/top_clean_sheets.csv → top 50 goalkeepers by clean sheets.
 - outputs/top_cts.csv → top 50 by Creative Threat Score (CTS).
 - outputs/top_sca_proxy.csv → top 50 by SCA proxy (total).
 - outputs/top_sca_proxy_per90.csv → top 50 by SCA proxy (per 90).
 - outputs/top_chance_creation.csv → top 50 by Chance Creation events.
 - outputs/sca.csv → raw SCA metric values for all players.

All scripts and outputs are version-controlled and reproducible.

Random seeds are documented (`np.random.seed(42)`) to ensure consistent results.

- **Appendix C:** Data lineage.

Source dataset: [Kaggle EPL Player Stats 2024/25](#).

Collected on: 2025-09-10.

Original format: CSV (`epl_player_stats_24_25.csv`).

Processing steps:

- Column renaming + cleaning.
- Derived metrics added (CTS, SCA proxy, Chance Creation).
- Filtered players with ≥ 900 minutes for per-90 comparisons.

Limitations:

- No granular xA (expected assists) → used proxies (fThird passes, crosses).
- Some players <300 minutes produce unstable per-90 rates.